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1. $f(z)=\frac{1}{2i}\left(\frac{z}{\bar z}-\frac{\bar z}{z}\right)(z\neq 0)$

设 $z=x+yi$

$f(z)=\frac{1}{2i}\left(\frac{x+yi}{x-yi}-\frac{x-yi}{x+yi}\right)$

$=\frac{1}{2i}\left(\frac{x^2-y^2+2xyi}{x^2+y^2}-\frac{x^2-y^2-2xyi}{x^2+y^2}\right)$

$=\frac{1}{2i}\cdot\frac{4xyi}{x^2+y^2}$

$=\frac{2xy}{x^2+y^2}$

$\lim\limits_{\substack{x\to 0\\ x=y}}=\frac{2y^2}{2y^2}=1$

$\lim\limits_{\substack{x=0\\ y\to 0}}=0\neq 1$

∴极限不存在

2. 解: 设 $z=x+yi$ $\quad |z|=\frac{|i+1|}{|w|}=\frac{\sqrt{2}}{|w|}$

$x^2+y^2=4$

$|z|^2=4 \qquad \frac{2}{|w|^2}=4$

$z=\frac{i+1}{w} \qquad |w|^2=\frac{1}{2}$

圆心为原点,半径为$\frac{\sqrt{2}}{2}$的圆

3. 解:

$\begin{cases}u(x,y)=x^3-y^3\\ v(x,y)=2x^2y^2\end{cases}$

其在复平面解析

若要使$f(z)$为可导函数,则

$u_x=v_y$

$3x^2=4x^2y$

$3=4y, y=\frac{3}{4}$ 或 $x=0$

$u_y=-v_x$

$-3y^2=4y^2x$

$y=0$ 或 $x=-\frac{3}{4}$

∵没有形成区域

∴不是解析函数

$f(z)=3x^2+4xy^2i$

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4. 解: $\oint_{|z|=1} \frac{|dz|}{|z-2|^2}$

$\left(|dz| = \frac{|z|dz}{iz} \right)$ ☆☆☆

$z = e^{i\theta}$　$dz = ie^{i\theta}d\theta$

$|dz| = |ie^{i\theta}d\theta| = d\theta = \frac{dz}{iz}$

$\therefore$ 原式 $= \oint_{|z|=1} \frac{\frac{dz}{iz}}{(z-2)(\bar{z}-2)}$

$= -i \oint_{|z|=1} \frac{dz}{z(z-2)(\bar{z}-2)}$

$= -i \oint_{|z|=1} \frac{dz}{z(1-2z-2\bar{z}+4)}$

$= -i \oint_{|z|=1} \frac{dz}{z(5-2z-\frac{2}{z})}$

$= -i \oint_{|z|=1} \frac{dz}{-2z^2+5z-2}$

$= i \oint_{|z|=1} \frac{dz}{2z^2-5z+2}$

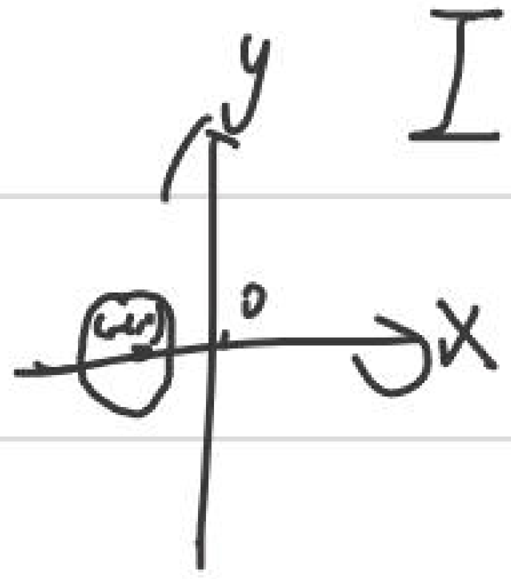
$= i \oint_{|z|=1} \frac{dz}{(z-2)(2z-1)}$

$\begin{matrix} 1 & -2 \\ 2 & -1 \end{matrix}$　$= i \cdot 2\pi i \left(\frac{1}{z-2} \right) \Big|_{z=\frac{1}{2}}$

$= -2\pi \cdot -\frac{2}{3}$

$= \frac{4\pi}{3}$

5. (1) $I_j = \oint_{C_j} \frac{z}{(z+1)(z-1)^2} dz \ (j=1,2,3)$



$I_1 = \oint_{|z+1|=\frac{1}{2}} \frac{z}{(z+1)(z-1)^2} dz$

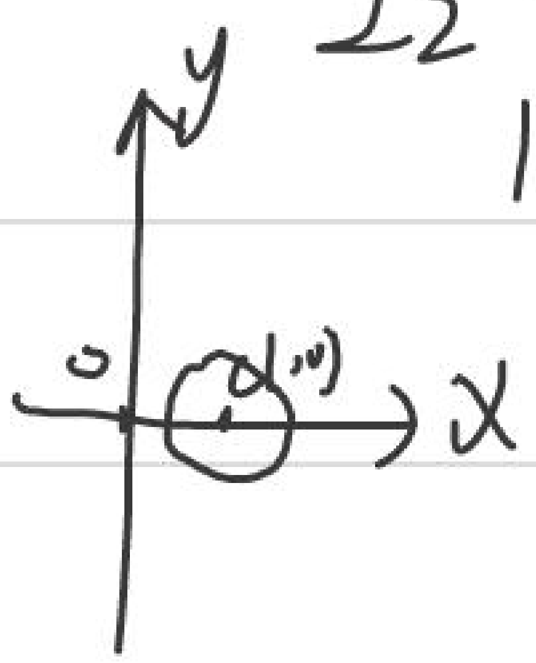
$= \oint_{|z+1|=\frac{1}{2}} \frac{\frac{z}{(z-1)^2}}{z+1} dz$

$= 2\pi i \cdot f(z_0=-1)$

$= 2\pi i \frac{-1}{4}$

$= -\frac{\pi i}{2}$

(2)



$I_2 = \oint_{|z-1|=\frac{1}{2}} \frac{z}{(z+1)(z-1)^2} dz$

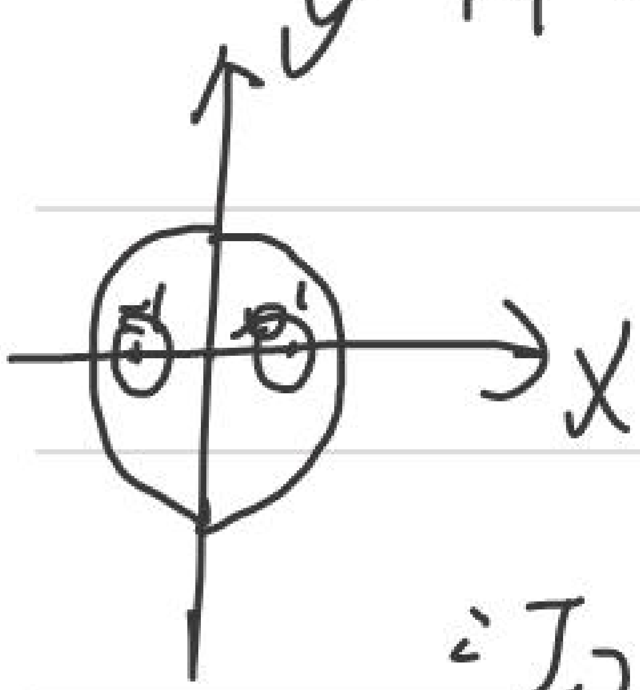
$= \oint_{|z-1|=\frac{1}{2}} \frac{\frac{z}{z+1}}{(z-1)^2} dz$

$= \frac{2\pi i}{1!} \left(\frac{z}{z+1} \right)' \Big|_{z=1}$

$= 2\pi i \cdot \frac{z+1-z}{(z+1)^2} \Big|_{z=1}$

$= 2\pi i \frac{1}{4} = \frac{\pi i}{2}$

(3) $I_3=\oint_{|z|=2}\frac{z}{(z+1)(z-1)^2}dz$



作C_1,C_2分别包含且仅含 $z_0=-1$, $z_1=1$ 的正向曲线

$$\therefore I_3=\oint_{C_1}\frac{\frac{z}{(z-1)^2}}{z+1}dz+\oint_{C_2}\frac{\frac{z}{(z+1)}}{(z-1)^2}dz$$

$$=0$$

6、证明:

$$\frac{R^2-|z|^2}{2\pi i}\oint_{|\eta|=R}\frac{f(\eta)}{(\eta-z)(R^2-\bar{z}\eta)}d\eta=f(z)$$

∵ $|z|<R$

∴ $|\bar{z}|<R$.

$$\therefore \oint_{|\eta|<R}\frac{f(\eta)}{(\eta-z)(R^2-\bar{z}\eta)}d\eta=2\pi i\frac{f(z)}{R^2-\bar{z}\cdot z}$$

$$=2\pi i\frac{f(z)}{R^2-|z|^2}$$

$$\therefore \frac{\cancel{R^2-|z|^2}}{\cancel{2\pi i}}\cdot\cancel{2\pi i}\frac{f(z)}{\cancel{R^2-|z|^2}}=f(z)$$

原式得证.

7. 解:

$u=2xy-2y$

∵ 解析函数

∴ $u_x=v_y=2y$

$v_y=2y$

$v=y^2+g(x)$

$\therefore f(z)=f(x+yi)=2xy-2y+(y^2+g(x))i$

$v_x=-u_y$

$g'(x)=2-2x$

$g(x)=2x-x^2+C$

$v=y^2+2x-x^2+C$

$\because f(2)=-i$

$v(2,0)=-1$

$\therefore C=-1$

$\therefore f(z)=2xy-2y+(y^2+2x-x^2-1)i$

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8. 解:

$$f(z)=z^5\left(\frac{A}{z-2}+\frac{B}{z-3}\right)$$

$$(A+B)z-(3A+2B)=1$$

$$\begin{cases}A+B=0\\ 3A+2B=-1\end{cases}$$

$$\begin{cases}A=-1\\ B=1\end{cases}$$

$$\therefore f(z)=z^5\left(\frac{-1}{z-2}+\frac{1}{z-3}\right)$$

$$=z^5\left(\frac{\frac{1}{2}}{1-\frac{z}{2}}+\frac{-\frac{1}{3}}{1-\frac{z}{3}}\right)$$

$$=z^5\left(\frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{z}{2}\right)^n+\left(-\frac{1}{3}\right)\sum_{n=0}^{\infty}\left(\frac{z}{3}\right)^n\right)$$

$$=z^5\left(\sum_{n=0}^{\infty}\frac{z^n}{2^{n+1}}-\sum_{n=0}^{\infty}\frac{z^n}{3^{n+1}}\right)$$

$$=\sum_{n=0}^{\infty}\frac{z^{n+5}}{2^{n+1}}-\sum_{n=0}^{\infty}\frac{z^{n+5}}{3^{n+1}}$$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{2^{n+1}}-\frac{1}{3^{n+1}}\right)z^{n+5}$$

$-2-n+1$

$=-n-1$

9. 解: $f(z)=\frac{1}{(z^2+1)^2}$

在 $z=i$ 展开

$$f(z)=\frac{1}{(z+i)^2(z-i)^2}$$

$$=\frac{1}{(z-i)^2}\cdot\frac{1}{(z+i)^2}$$

$$=\frac{1}{(z-i)^2}\cdot(z-i+2i)^{-2}$$

$$=\frac{1}{(z-i)^2}\cdot\left(\frac{z-i}{2i}+1\right)^{-2}\cdot(2i)^{-2}$$

$$=\frac{1}{-4(z-i)^2}\left(1+\frac{z-i}{2i}\right)^{-2}$$

$\left|\frac{z-i}{2i}\right|<1$ 上解析

即 $|z-i|<2$ 上解析

$$\therefore 原式=\frac{1}{-4(z-i)^2}\cdot\left[1+\sum_{n=1}^{\infty}\frac{-2(-3)\cdots(-2-(n-1))}{n!}\left(\frac{z-i}{2i}\right)^n\right]$$

$$=\frac{1}{-4(z-i)^2}\left[1+\sum_{n=1}^{\infty}\frac{(n+1)!(-1)^n}{n!}\left(\frac{z-i}{2i}\right)^n\right]$$

$$=\frac{1}{-4(z-i)^2}+\left(-\frac{1}{4}\right)\sum_{n=1}^{\infty}(n+1)(-1)^n\frac{(z-i)^{n-2}}{(2i)^n}$$

$$C_{-1} = \frac{-\frac{1}{4} \times 2 \times (-1)}{2i}$$

$$= \frac{\frac{1}{2}}{2i}$$

$$= -\frac{1}{4} i$$

$\therefore$ 积分 $= 2\pi i \cdot -\frac{1}{4} i$

$$= -\frac{\pi}{2}(-1) = \frac{\pi}{2}$$

10 解：取 $r = \frac{1}{2}$

$$I = 2\pi i \{ Res[f(\tfrac{1}{s})(\tfrac{1}{s^2}), 0] + Res[f(\tfrac{1}{s})(\tfrac{1}{s^2}), \tfrac{1}{4}]$$

$$= 2\pi i \{ \frac{1}{-1} \lim_{s \to 0} \cos\frac{1}{(\frac{1}{s}-1)^2} + \frac{\frac{15}{16} \times \frac{63}{64} \times \cos\frac{1}{9}}{\frac{1}{4} \times (\frac{1}{512}-1)}$$

$$= 2\pi i \left(-1 + \frac{15 \times 63 \cos\frac{1}{9}}{\frac{1}{2} - \frac{1024}{4}}\right) = 2\pi i \left(-1 + \frac{15 \times 63 \cos\frac{1}{9}}{\frac{1}{2} - 256}\right) = ?$$

$-2\pi i$

$$\frac{\left(\frac{1}{s^2} - 2\right)\left(\frac{1}{s^3} - 1\right) \cos\frac{1}{(\frac{1}{s}-1)^2}}{\left(2 - \frac{1}{s^5}\right)\left(\frac{1}{s} - 4\right)} \cdot \frac{1}{s^2}$$

$$= \frac{\cancel{s}(1-s^2)(1-s^3)\cos\frac{1}{(\frac{1}{s}-1)^2}}{(2s^5-1)(1-4s)} \cdot \frac{1}{s^4}$$

$|s| = \frac{1}{3}$

$s^5 = \frac{1}{2}$　$s = \frac{1}{4}$

$$\frac{(1-s^4)(1-s^3)\cos\frac{1}{(\frac{1}{s}-1)^2}}{s(2s^5-1)(1-4s)}$$

$\frac{1}{512} - 1$

$2 \times (\frac{1}{4})^5 - 1$

$64 \times 16 = 384 + 64 = 1024$

$\frac{}{s^5}$

11. $\sum_{-\infty}^{+\infty} \frac{e^{ix}}{(x^2+4)(x-1)} dx$ $\therefore \sum_{-\infty}^{+\infty} \frac{\sin x}{(x^2+4)(x-1)} dx$

$= \sum_{-\infty}^{\infty} \frac{e^{iz}}{(z^2+4)(z-1)} dz$ $= \underline{I}_m \left(\sum_{-\infty}^{+\infty} \frac{e^{ix}}{(x^2+4)(x-1)} dx \right)$

$z_{0,1} = \pm 2i$, $z_2 = 1$ $= -\frac{\pi e^{-2}}{5}$

$= 2\pi i \, \text{Res}\left[\frac{e^{iz}}{(z^2+4)(z-1)}, 2i\right]$

$= 2\pi i \cdot \frac{e^{-2}}{(z+2i)(z-1)}\Big|_{z=2i}$

$= 2\pi i \cdot \frac{e^{-2}}{4i(2i-1)}$

$= \frac{\pi e^{-2}}{2(2i-1)} = \frac{\pi e^{-2}}{2} \frac{1}{2i-1} = \frac{-\pi e^{-2}}{2} \frac{1}{1-2i}$

$= \frac{-\pi e^{-2}}{2} \frac{1+2i}{5}$

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